

# PROJECT

## XONIX GAME



*“Greatness isn’t a gift, it’s a grind. Chase your goals with fire, break through barriers and grow through every challenge. The top is reserved for those who never back down.”*

| ASSIGNING DATE | DUE DATE                  | TOTAL MARKS     |
|----------------|---------------------------|-----------------|
| July 25, 2026  | August 6, 2026 – 11:59 PM | 120 (+10 Bonus) |

*A group activity - 2 members per group*

## INSTRUCTIONS & SUBMISSION GUIDELINES

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### ACADEMIC INTEGRITY

Plagiarism is strongly forbidden. If a group copies from another (with or without knowledge), both groups receive a zero and face Disciplinary Committee action, which can extend to an F in the course.

**Any use of AI or plagiarism in the report results in a zero on the project or an F in the course. Using an LLM to rephrase or paraphrase is considered an AI-generated document.**

### Submission Rules

- This is a group activity of 2 people.
- Submissions other than Google Classroom (e.g. email) will not be accepted and no late submissions will be accepted.
- The group leader is responsible for correct and timely submission; no relaxation will be given to anyone.
- Combine all your work into a single .zip file, named ROLL-NUM.zip (e.g. 23i0000.zip) and submit it on Google Classroom.

### Coding & Report Guidelines

- You may use anything to code this project, though the contents covered in the course.
- You may take implementation ideas from internet resources, but do not copy any chunk of code exactly: zero tolerance for plagiarism.
- Divide and conquer: break the project into manageable subtasks across the days available.
- Use your imagination, an excellent, appealing solution can earn bonus marks.
- Follow good programming practice: well-commented, well-indented code with meaningful variable names.
- For unspecified details, use your creativity to design solutions aligned with the project's goals and document your rationale in the report.

## PROJECT BRIEF

You are working as a game developer when a client approaches you with an idea for a game. Some features have already been implemented, while others are yet to be developed.

### Basic Game Rules (Already Implemented)

- A player can move in any of the four directions: right, left, up and down.
- While building tiles, the player cannot reverse their movement. e.g. if moving upward while building, moving directly downward will cause the player's death.
- If an enemy touches a tile that is under construction, the player will die.

### Baseline Starter Code

Base line started code is provided in a zip format including guidelines and commands for installation and build project.

## FEATURES TO IMPLEMENT

*The following features must be implemented and are graded as shown. Bonus marks are only awarded after all required features are complete.*

### 1. Basic Features

**10 MARKS**

- **Game modes:** single-player and two-player modes.
- **Start menu:** Start Game, Select Level and Scoreboard.
- **End menu:** end the game properly and show the final score.
- **High-score feedback:** highlight the result when a new high score is achieved.
- **End options:** Restart, Main Menu and Exit Game.

### 2. Difficulty & Enemy Count

**20 MARKS**

Allow the player to select from the following modes:

- **Easy:** 2 enemies.
- **Medium:** 4 enemies.
- **Hard:** 6 enemies.
- **Continuous Mode:** start with 2 enemies and add 2 more every 20 seconds throughout the game.

### 3. Movement Counter

**5 MARKS**

- **Track moves:** display the total number of moves made by the player.
- **Move definition:** count one move each time the player starts building tiles.

### 4. Enemy Speed & Movement

**20 MARKS**

- **Elapsed time:** track and display the game duration.
- **Speed scaling:** increase enemy movement speed by a fixed amount every 20 seconds.
- **Pattern switch:** after 30 seconds, half of the enemies must change from linear movement to defined geometric patterns.
  - **Non-random paths:** implement at least two specific geometric patterns, such as zig-zag and circular movement.
  - **Separate functions:** implement each pattern in its own clearly labelled function.

### 5. Scoring & Reward System

**10 MARKS**

- **Base score:** capturing one tile grants 1 point.
- **Double bonus:** capturing more than 10 tiles in a single move awards x2 points.
- **Threshold reduction:** after 3 bonus occurrences, reduce the threshold from more than 10 tiles to more than 5 tiles.
- **Quadruple bonus:** after 5 total bonus occurrences, capturing more than 5 tiles awards x4 points.
- **First power-up:** at score 50, award a power-up that freezes all enemies for 3 seconds.
- **Additional power-ups:** award at 70, 100, 130 and every additional 30 points.
- **Inventory:** unused power-ups remain available and stack.

### 6. Scoreboard

**10 MARKS**

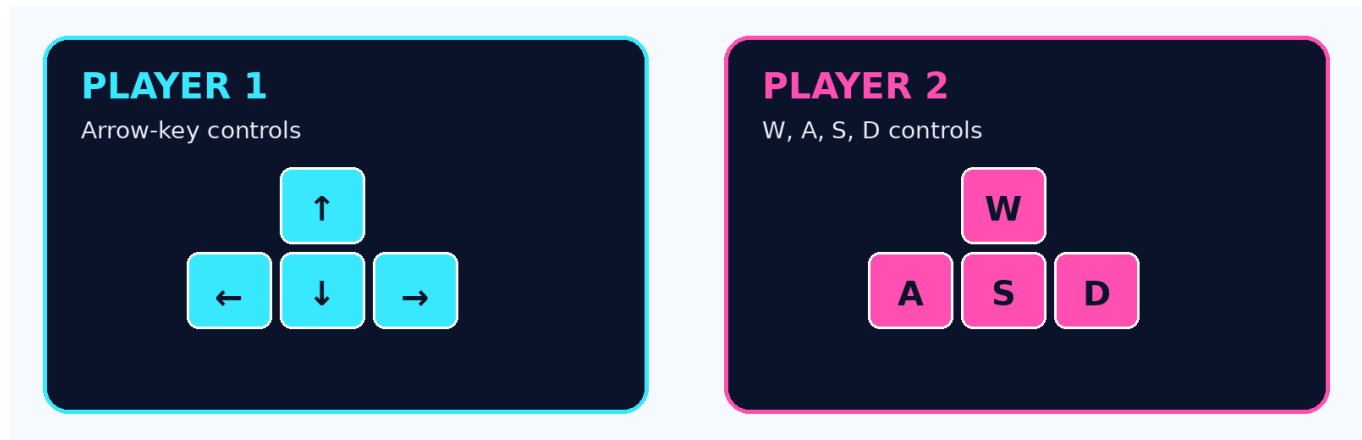
- **Storage:** use a text file (.txt) to maintain the top 5 scores.
- **Ordering:** store scores in descending order.
- **Record details:** include both the score and the time taken.
- **Automatic update:** on game over, update the file if the result belongs in the top 5.

### 7. Two Player Mode Rules

**25 MARKS**

- **Shared board:** both players play on the same game board.
- **Shared time:** both players use one game timer.
- **Individual information:** display scores and power-ups on each player's respective side of the screen.

- **Feature parity:** all general game features apply to both players.



- Since both players share the board, they can interact. If a player touches the other player or their in progress tiles:
  - If both players collide while constructing tiles, both players die.
  - If Player 1 touches Player 2's constructing tile, Player 1 dies and vice versa.
  - If Player 1 collides with Player 2 while only Player 1 is constructing tiles, Player 1 dies and vice versa.
- Power-up usage in two-player mode stops both the enemies and the other player for 3 seconds.
- When both players are eliminated, the game ends and the winner is determined by the highest score.

## 8. Report

**20 MARKS**

A detailed report of 1-2 pages is required, covering:

- The overall project workflow and game development approach.
- A workflow diagram (e.g. using draw.io).
- Task distribution between both team members.
- Implementation progress of each feature against the requirements above.
- Document important choices made for unspecified details.

## 9. Bonus Features

**10 MARKS**

- Add sound effects for events such as capturing tiles, achieving bonuses, or activating power-ups.
- Change the background color from black to a suitable alternative when a power-up is used.

**Note: Bonus marks will only be awarded after completing all required project features.**

## FEATURE CHECKLIST

| Feature                   | Marks      | Status | Notes                               |
|---------------------------|------------|--------|-------------------------------------|
| 1. Basic Features         | 10         |        | Single/Two Player, Menus            |
| 2. Difficulty & Enemies   | 20         |        | Easy / Medium / Hard / Continuous   |
| 3. Movement Counter       | 5          |        | Track tile-building moves           |
| 4. Enemy Speed & Patterns | 20         |        | Speed increase + geometric paths    |
| 5. Scoring & Rewards      | 10         |        | Points, bonuses, power-ups          |
| 6. Scoreboard             | 10         |        | Top 5 scores in .txt file           |
| 7. Two-Player Mode        | 25         |        | Shared board, collisions, power-ups |
| 8. Report                 | 20         |        | Workflow, diagrams, tasks           |
| 9. Bonus Features         | 10         |        | Sound effects, background color     |
| <b>Total Marks</b>        | <b>120</b> |        | <b>+ 10 Bonus</b>                   |

## LIBRARIES

### ✗ RESTRICTED

- <vector>
- <algorithm>
- Built-in lists
- Built-in queues
- Built-in stack

### ✓ ALLOWED

- <SFML/Graphics.hpp>
- <time.h>
- <cmath>
- <cstdlib>
- <string>
- <cstring>

# GOOD LUCK

*Plan carefully. Build honestly. Test thoroughly. Explain confidently.*